

Guide 31 - Plumber, Pipefitter, and Plumbing Technician

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Purpose: Practical career-entry and advancement guide for plumbers, pipefitters, steamfitters, plumbing technicians, and closely related piping-service roles.

This guide is educational. It does not guarantee employment, wages, admission, funding, reimbursement, apprenticeship placement, certification, licensing, promotion, or legal authority to perform plumbing, gas, pressure-piping, fire-protection, medical-gas, or other regulated work. Requirements vary by jurisdiction, system, employer, and task. Verify current rules before paying for training or performing hazardous or regulated work.

1. What this work is and who it may fit

Plumbers, pipefitters, and steamfitters install, test, maintain, troubleshoot, and repair piping, fixtures, valves, equipment connections, and related systems. Depending on the occupation and jurisdiction, systems may carry potable water, wastewater, heating water, steam, compressed air, chemicals, fuel gas, process fluids, medical gases, fire-protection water, or other substances.

This field can fit people who enjoy practical problem-solving, measurement, tools, diagrams, construction, mechanical systems, troubleshooting, customer service, and work that produces visible results. It also requires disciplined safety judgment. Piping work can involve pressure, stored energy, hot surfaces, chemicals, sewage, biological hazards, fuel gas, combustion, sharp tools, lifting, ladders, trenches, confined spaces, hot work, and interaction with electrical or mechanical equipment.

A job title alone does not authorize every piping task. A residential plumber, industrial pipefitter, steamfitter, gasfitter, sprinklerfitter, backflow technician, medical-gas installer, facilities technician, and plumbing helper may have different training, licensing, certification, code, and supervision requirements.

2. What workers actually do

Typical duties may include:

- reading drawings, specifications, piping diagrams, fixture schedules, equipment manuals, work orders, and local code requirements;
- measuring, cutting, joining, supporting, and routing approved piping materials within authorized scope;
- installing or replacing fixtures, valves, traps, drains, vents, fittings, pumps, water heaters, and other approved components;
- inspecting systems for leaks, blockage, corrosion, damage, improper support, abnormal pressure, or other defects;

- testing installed systems using approved procedures, instruments, and pressure limits;
- troubleshooting drainage, water-supply, hydronic, process, or other piping problems within verified competence;
- maintaining records of inspections, tests, repairs, permits, materials, unresolved conditions, and required escalation;
- coordinating with inspectors, engineers, contractors, electricians, HVAC technicians, building operators, utility personnel, and other trades;
- performing preventive maintenance and emergency response within legal and employer-authorized scope;
- stopping when the task requires a license, permit, certification, competent person, qualified supervision, confined-space program, excavation protection, hot-work controls, or other authority not in place.

Title distinctions matter

Plumbers commonly work on water, drainage, fixtures, sanitary systems, and building plumbing.

Pipefitters and steamfitters often work on industrial/commercial piping, process systems, heating/cooling piping, chemicals, gases, and higher-pressure systems.

Gasfitters and sprinklerfitters may be separate regulated specialties.

Plumbing technician is an employer-dependent title that may describe helper, maintenance, facilities, service, installation, or more advanced duties.

Do not assume that one title's wage, license, or authority applies to another.

3. Safety, pressure, hazardous energy, and task boundaries

Plumbing and piping work can expose workers to multiple energy sources and hazardous environments.

Hazardous energy

OSHA 29 CFR 1910.147 addresses covered servicing and maintenance where unexpected energization, startup, or release of stored energy could cause injury. Relevant energy may be electrical, mechanical, hydraulic, pneumatic, chemical, thermal, pressure-related, or otherwise stored.

Do not assume that a closed valve, stopped pump, switch position, pressure gauge, or software command makes a system safe. Follow the employer's applicable isolation procedure, identify all energy sources, control or relieve stored energy, verify isolation, and remain within authorized-worker boundaries.

Pressure and hot systems

Pressurized water, steam, compressed air, chemicals, fuel gas, process fluids, or hot hydronic systems can release energy suddenly. Never loosen a fitting, open a flange, remove a valve, cut a pipe, or begin hot work because a gauge “looks safe” unless the approved procedure confirms isolation, depressurization, temperature, contents, and authorization.

Confined spaces

OSHA 1926.1201 identifies construction confined-space examples that directly overlap plumbing and piping work, including manholes, sewers, storm drains, water mains, tanks, pits, lift stations, cesspools, and boilers. Entry may require atmospheric testing, ventilation, attendants, permits, rescue planning, communication, and other controls.

Excavations and trenches

OSHA Subpart P applies to excavations, including trenches. OSHA 1926.651 requires underground installations such as sewer, fuel, electric, and water lines to be located before excavation and requires contact with utility owners as specified.

Never enter or work around an excavation based only on visual confidence. Protective systems, competent-person inspections, access/egress, spoil placement, atmosphere, water accumulation, traffic, utilities, and other hazards may apply.

Stop and escalate when

- system contents, pressure, temperature, energy sources, isolation points, or utility locations are uncertain;
- required permits, drawings, test instruments, PPE, protective systems, ventilation, gas testing, or rescue provisions are missing;
- fuel-gas, medical-gas, fire-protection, backflow, high-pressure, steam, chemical, hot-work, or specialized piping is outside your verified scope;
- an excavation or confined-space program is not clearly controlled by the responsible employer/competent person;
- a customer or supervisor requests bypassing a required safety device, code, inspection, or permit;
- the work requires a license, certification, trade status, or supervision you do not have;
- contamination, sewage, chemical, biological, asbestos, lead, mold, or other hazardous-material concerns exceed the established procedure.

4. United States licensing, codes, and legal-work boundaries

The United States does not have one national plumbing license that authorizes independent plumbing everywhere. BLS states that most states and some localities require plumbers to be licensed. Some jurisdictions also license pipefitters or steamfitters and may require special authorization for gas-line work.

Licensing commonly depends on a combination of supervised experience, apprenticeship or training, examinations, and jurisdiction-specific requirements. Categories may include apprentice or trainee, journey-level plumber, master plumber, plumbing contractor, gasfitter, specialty classifications, or other roles.

Before paying for training, verify:

- the actual licensing or registration authority where you intend to work;
- required supervised hours and who may verify them;

- classroom or technical-instruction requirements;
- apprenticeship or trainee registration rules;
- examination eligibility and content;
- journey/master/contractor distinctions;
- adopted plumbing, mechanical, fuel-gas, building, fire, and other codes;
- permit and inspection requirements;
- continuing education and renewal;
- reciprocity, if any;
- separate qualifications for gas, medical gas, backflow, fire protection, pressure piping, welding/brazing, or other specialties.

A school certificate does not itself prove legal authorization to perform independent plumbing work.

5. Entry routes and transferable skills

BLS identifies a high school diploma or equivalent as typical entry education and apprenticeship as the typical on-the-job training route for plumbers, pipefitters, and steamfitters.

Common routes include:

- paid apprenticeship programs sponsored by unions, trade associations, or employers;
- helper or trainee positions that lead into structured apprenticeship;
- pre-apprenticeship programs connected to legitimate employers or apprenticeship systems;
- public technical schools or community colleges;
- military, facilities, construction, HVAC, welding, maintenance, utilities, or industrial experience translated carefully into civilian requirements;
- employer-supported training with progressive responsibility;
- public vocational training such as SENA hydrosanitary programs in Colombia.

Useful transferable skills include:

- measurement and applied mathematics;
- reading blueprints and piping diagrams;
- safe hand- and power-tool use;
- mechanical reasoning;
- documentation and work-order discipline;
- customer service;
- teamwork across trades;
- material identification;
- troubleshooting based on evidence;
- willingness to stop when the scope or system condition is uncertain.

Do not claim journey, master, gasfitter, backflow, medical-gas, welding, Red Seal, contractor, or licensing status you do not actually hold.

6. United States labor-market evidence

Official national statistics

BLS reports for Plumbers, Pipefitters, and Steamfitters (SOC 47-2152):

- **2024 employment: 504,500;**
- **May 2024 median pay: \$62,970 per year / \$30.27 per hour;**
- **lowest 10 percent: less than \$40,670;**
- **highest 10 percent: more than \$105,150;**
- **projected 2034 employment: 527,200;**
- **projected growth, 2024-2034: 4%;**
- **projected numeric growth: 22,700;**
- **average projected openings: about 44,000 per year over the decade.**

These figures cover the combined occupation. They are not plumber-only, pipefitter-only, apprentice, local-market, or guaranteed wage figures.

Current non-government plumber estimate

Indeed's U.S. Plumber salary page, updated August 2, 2026, reported an average base salary of approximately 30.53 *per hour*, with displayed low/high values around ***16.77-\$55.59 per hour** and an overtime indicator.

This is a **non-government plumber-title estimate** based on Indeed's methodology and is not definition-equivalent to BLS SOC 47-2152. Use it only as a current market cross-check.

When comparing offers, ask about apprentice wage progression, overtime, on-call duty, travel, tools, PPE, licensing fees, training time, union dues, vehicle use, testing costs, and benefits.

7. Free and low-cost U.S. pathways and funding

Apprenticeship first where appropriate

Because apprenticeship is the typical training route, compare legitimate paid apprenticeship and employer-supported options before assuming private-school debt.

FAFSA and federal student aid

Current 2026–27 Federal Student Aid guidance states that eligible students at participating college, career-school, or trade-school programs may access federal grants, Federal Work-Study, and federal loans through the FAFSA process.

FAFSA submission is free. Do not assume a particular private plumbing program is eligible. Verify the institution, program, student eligibility, costs, outcomes, and whether the training satisfies licensing/apprenticeship requirements.

WIOA and American Job Centers

U.S. Department of Labor WIOA programs provide career and training services, including classroom and work-based learning. American Job Center services and eligibility vary locally.

Ask about:

- plumbing/construction-trade training on current approved-provider lists;
- pre-apprenticeship or apprenticeship referral;
- on-the-job training;
- supportive services such as transportation, tools, childcare, or PPE when locally available;
- accommodations and accessible training;
- whether written authorization is required before enrollment.

Employer educational assistance

IRS guidance dated April 20, 2026 states that under a qualifying Section 127 employer educational-assistance plan, up to **\$5,250** of qualifying benefits may be excluded from an employee's gross income for calendar year 2026, subject to the tax rules and employer plan.

This does not mean every employer offers \$5,250 or that every course qualifies. Obtain written terms before enrolling.

8. Apprenticeship and work-based learning

BLS states that most plumbers, pipefitters, and steamfitters learn through a **4- or 5-year apprenticeship**. Apprentices typically receive about **2,000 hours of paid on-the-job training each year** plus technical instruction in subjects such as safety, local codes, blueprint reading, mathematics, applied physics, and chemistry.

Apprenticeship.gov remains a current federal locator for approved occupations, registered partners, and job opportunities. However, the current public web fetch did not expose a reliable occupation-specific approval result for O*NET 47-2152.00 because the detailed listing depends on client-side content.

Therefore:

- it is accurate to say apprenticeship is the typical pathway because BLS says so;
- do not make a new blanket claim that the exact O*NET code is currently approved for Registered Apprenticeship unless directly verified;
- verify the actual sponsor, registered occupation/code, registration authority, paid status, starting wage, wage progression, work hours, related instruction, licensing credit, completion credential, tools/PPE, tuition, and probation/cancellation terms.

A helper job, internship, pre-apprenticeship, school practicum, or employer training program can be useful without being a Registered Apprenticeship.

9. Canada pathway

Occupation and wages

Government of Canada Job Bank maps plumber-related work to **NOC 72300**. National wage data updated November 19, 2025 report:

- **C\$21.00/hour low;**
- **C\$34.00/hour median;**
- **C\$46.00/hour high.**

These are national reference values, not guaranteed offers.

Red Seal — Plumber

Red Seal identifies **Plumber** as NOC 72300 and a designated Red Seal trade in participating provinces and territories. The national trade description spans water, hydronic, drainage/waste/vent, low-pressure steam, residential fire, chemical, irrigation, medical-gas, process-piping, compressed-air, water-conditioning, fuel-piping, sewage/water-treatment, and related systems.

That broad description does not mean every apprentice or plumber is automatically authorized for every specialty. Province/territory requirements may govern apprenticeship registration, compulsory-trade status, certification, gas, medical gas, backflow, contractor status, permits, inspections, and other tasks.

Verify the apprenticeship authority and trade rules where you will train and work.

10. Colombia pathway

SENA free-first options

Current SENA Betowa evidence includes:

- **Instalación de redes hidrosanitarias** — complementary, in-person, **48 hours**;
- **Instalación de redes hidrosanitarias en PVC y griferías** — complementary, in-person, **48 hours**;
- **Básico de instalaciones sanitarias domiciliarias** — complementary, in-person, **60 hours**;
- **Construcción de edificaciones** — titled **Técnico, 2,160 hours**, with hydrosanitary installation among broader construction competencies.

These can be useful public/free-first learning routes. They are not interchangeable:

- a 48- or 60-hour complementary course is a short skills course, not equivalent to a 4- or 5-year U.S. apprenticeship;
- the 2,160-hour Construction Technician program is broader than plumbing and should be treated as an adjacent foundation rather than an international plumber credential;

- cohorts, cities, schedules, modality, admission, and availability change and must be rechecked in Betowa.

The controlled research did not establish one national Colombian plumber license that can safely be generalized to every plumbing, gas, pressure, fire-protection, utility, or building-system task. Verify current technical regulations, local permits/inspections, public-utility requirements, employer/contractor conditions, gas-system authorization, and any competency or certification rules that apply to the actual work.

11. Latin America pathway

Plumbing titles, vocational programs, codes, gas rules, public-health requirements, utility regulations, licensing, permits, and inspection systems differ by country and often by locality.

For any country:

1. identify the authority responsible for plumbing/building/gas regulation;
2. check official vocational-training and public-employment systems;
3. identify the adopted plumbing, sanitary, mechanical, fuel-gas, building, and fire rules relevant to the task;
4. verify apprentice, trainee, technician, contractor, or professional categories;
5. confirm recognition of foreign education and experience;
6. compare public/free training before paying a private provider;
7. verify current employer demand and local wages separately.

Do not transplant U.S. OSHA/licensing, Canadian Red Seal, or Colombian SENA rules into another jurisdiction.

12. Skills to build deliberately

Technical foundations

Develop competence in:

- measurement, fractions, ratios, slope, basic geometry, and unit conversion;
- reading blueprints, isometrics, piping diagrams, fixture schedules, and specifications;
- pipe, tubing, fittings, valves, supports, fixtures, drains, vents, and material identification;
- water-supply and drainage concepts;
- safe joining methods appropriate to material, training, and authorization;
- testing and leak-detection concepts within approved procedures;
- troubleshooting based on measurements and observations;
- preventive maintenance and inspection;
- safe tool use;
- documentation and digital work-order systems.

Advanced or specialty topics—fuel gas, medical gas, backflow, fire protection, steam, high pressure, industrial process piping, welding/brazing, water treatment, and cross-connection

control—may require additional authorization and should not be treated as beginner-independent work.

Professional skills

Build:

- customer communication;
- accurate estimates and documentation within role scope;
- teamwork across trades;
- cleanliness and contamination-control discipline;
- punctuality and reliability;
- respect for permits, inspections, and supervision;
- willingness to disclose uncertainty rather than hide it.

13. Digital systems, cybersecurity, and privacy

Modern plumbing and piping work may involve building-management systems, smart leak detectors, pumps, water-treatment controls, digital pressure/flow sensors, connected water heaters, access-control systems, maintenance platforms, remote diagnostics, and customer records.

Protect digital and physical information:

- use only approved devices, applications, accounts, and diagnostic tools;
- do not connect personal devices to customer or industrial networks without authorization;
- protect credentials and MFA methods;
- do not publish building plans, utility layouts, security information, customer addresses, access codes, proprietary process diagrams, or credentials;
- verify unusual remote-control or configuration requests;
- escalate suspected compromise or abnormal control behavior to responsible IT/OT, facilities, controls, engineering, or security teams.

Cybersecurity knowledge can be useful in smart buildings, facilities, utilities, water systems, and industrial environments, but it does not replace trade qualifications.

14. Responsible use of artificial intelligence

AI can help with low-risk study and administrative practice using public or fictitious information.

Useful lower-risk examples include:

- explaining terminology;
- generating study questions;
- practicing interviews;
- creating learning plans;

- practicing unit conversions with fictional examples;
- organizing questions for a licensing board, apprenticeship sponsor, employer, Red Seal authority, SENA, or school;
- improving grammar in a fictitious work note.

Do not rely on AI as sole authority for real-world:

- fuel-gas or medical-gas work;
- pressure-system isolation or testing;
- hot tapping;
- backflow/cross-connection decisions;
- potable-water contamination control;
- pipe sizing or code compliance;
- trench or confined-space entry;
- hazardous-energy isolation;
- welding/brazing/hot work;
- permit or inspection decisions;
- licensing or credential scope.

AI can hallucinate code sections, pressure limits, material compatibility, formulas, manufacturer requirements, and legal exceptions. Verify real decisions against current adopted codes/standards, manufacturer information, employer procedures, competent/qualified supervision, and the authority having jurisdiction.

15. Accessibility and accommodations

Plumbing and piping training can involve physical, sensory, cognitive, communication, and environmental demands. Many people with disabilities can succeed with appropriate accommodations, task design, assistive technology, or specialization.

Helpful strategies may include:

- written checklists and numbered procedures;
- diagrams paired with plain-language descriptions;
- accessible digital training materials;
- speech-to-text or text-to-speech for documentation when safe and permitted;
- ergonomic tools or approved lifting aids;
- structured tool/material organization;
- additional time or accessible formats for written learning/exams when available;
- clear task sequencing and supervised practice.

Accommodation does not remove essential safety requirements or authorize work outside legal scope. Discuss needs with the training provider, employer, apprenticeship program, examination authority, or accommodation office.

16. Job search, portfolio, and interview preparation

Resume evidence

Emphasize truthful evidence of:

- supervised plumbing/piping training or apprenticeship status;
- licenses, registrations, certifications, or trainee cards actually held;
- blueprint/piping-diagram reading;
- tools and materials actually used;
- installation, maintenance, or repair work within verified scope;
- safety training, LOTO, confined-space, excavation, hot-work, or other credentials actually completed;
- work-order documentation and customer service;
- construction, HVAC, facilities, welding, maintenance, industrial, or utility experience relevant to the target role.

Never claim licensed plumber, journey/master status, gasfitter, Red Seal, medical-gas, backflow, welding, contractor, or pressure-piping competence unless supported.

Safe portfolio ideas

Use only safe, permitted, nonconfidential evidence:

- legitimate certificates/transcripts;
- classroom or training-board projects;
- fictional piping diagrams;
- public-manufacturer-based maintenance checklists;
- mock troubleshooting exercises;
- permitted photographs of deenergized/nonhazardous training projects with no customer or security information;
- a written example showing when you would stop and escalate.

Do not create portfolio material by performing unlicensed work, entering trenches/confined spaces for photos, opening pressurized systems, or exposing customer/building information.

Interview scenarios

Practice questions such as:

- How do you verify a piping system is safe before opening it?
- What would make you stop a job and call a supervisor?
- How do you distinguish plumbing work from gas or specialized piping work?
- What is your current apprentice, trainee, journey, or licensing status?
- How would you handle a suspected confined-space or trench hazard?
- How do you protect potable-water systems from contamination?
- How do you document an unresolved leak or defect?

- How do you protect customer information when using digital tools?

17. Twelve-week entry or advancement plan

Weeks 1-2 — Define the exact target

- Read the BLS occupation profile and current job postings.
- Decide whether your target is plumbing, industrial pipefitting, steamfitting, facilities plumbing, service, helper, apprenticeship, or another adjacent path.
- Identify the licensing or apprenticeship authority where you intend to work.
- In Canada, identify province/territory apprenticeship requirements.
- In Colombia, review current SENA Betowa offerings and local regulatory requirements.

Weeks 3-4 — Build safe fundamentals

- Review measurement, drawings, materials, drainage/water concepts, and safety terminology.
- Practice only in instructor-controlled or clearly safe training environments.
- Learn hazardous-energy, trench, confined-space, pressure, sanitation, and contamination-control concepts.

Weeks 5-6 — Compare pathways

- Compare paid apprenticeships, public technical programs, community colleges, employer training, SENA, union/industry programs, and pre-apprenticeships.
- Verify whether training counts toward licensing or apprenticeship requirements.
- Compare tuition, books, tools, PPE, travel, test fees, refund rules, and recognized hours.

Weeks 7-8 — Prepare job-search evidence

- Build a truthful resume.
- Gather legitimate records and references.
- Practice safety and scope-boundary interviews.
- Track applications and recurring requirements.

Weeks 9-10 — Verify funding and terms

Check:

- apprentice/trainee wages and progression;
- tuition and books;
- tools and PPE;
- exam/licensing fees;
- FAFSA eligibility when relevant;
- WIOA/American Job Center support;
- scholarships;
- employer Section 127 or other education assistance;
- repayment clauses;
- transportation/on-call expectations;

- accommodation processes.

Weeks 11-12 — Apply and refine

- Apply to roles matching your actual scope and training.
- Address one verified gap at a time.
- Recheck licensing/apprenticeship facts before major spending.
- Avoid accumulating unrelated certificates solely for appearance.

18. Verification checklist, sources, authorship, and license

Verify before enrolling or accepting work

- ☐ exact duties and system type;
- ☐ required apprentice/trainee/journey/master/contractor/specialty status;
- ☐ licensing and permit authority;
- ☐ required supervised hours and accepted training;
- ☐ applicable plumbing, sanitary, mechanical, fuel-gas, fire, building, and specialty codes;
- ☐ hazardous-energy, pressure, hot-work, trench, confined-space, contamination, and PPE procedures;
- ☐ gas, medical-gas, backflow, fire-protection, welding/brazing, or pressure-piping qualifications where relevant;
- ☐ school/program recognition and total cost;
- ☐ written funding/reimbursement eligibility;
- ☐ wages, progression, overtime, on-call, travel, and benefits;
- ☐ current Canada province/territory rules if applicable;
- ☐ current SENA availability and Colombian local requirements if applicable;
- ☐ cybersecurity/AI policy;
- ☐ accommodation process.

Controlled sources reviewed for this edition

United States occupation and market

- U.S. Bureau of Labor Statistics — Plumbers, Pipefitters, and Steamfitters: <https://www.bls.gov/ooh/construction-and-extraction/plumbers-pipefitters-and-steamfitters.htm>
- Indeed — Plumber salary (non-government): <https://www.indeed.com/career/plumber/salaries>

Apprenticeship, funding, and workforce support

- Apprenticeship.gov — Occupation Finder: <https://www.apprenticeship.gov/apprenticeship-occupations>
- Apprenticeship.gov — Job Finder: <https://www.apprenticeship.gov/apprenticeship-job-finder>

- Federal Student Aid — FAFSA 2026–27 steps: <https://studentaid.gov/articles/fafsa-student-steps/>
- U.S. Department of Labor — WIOA programs: <https://www.dol.gov/agencies/eta/wioa/programs>
- U.S. Department of Labor — American Job Centers: <https://www.dol.gov/agencies/eta/american-job-centers>
- IRS — Section 127 educational assistance, April 2026: <https://www.irs.gov/newsroom/irs-updates-frequently-asked-questions-about-section-127-educational-assistance-programs>

Safety

- OSHA — 1910.147 hazardous energy: <https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.147>
- OSHA — Confined Spaces in Construction 1926.1201: <https://www.osha.gov/laws-regs/regulations/standardnumber/1926/1926.1201>
- OSHA — Excavations 1926.650: <https://www.osha.gov/laws-regs/regulations/standardnumber/1926/1926.650>
- OSHA — Specific Excavation Requirements 1926.651: <https://www.osha.gov/laws-regs/regulations/standardnumber/1926/1926.651>

Canada

- Government of Canada Job Bank — Plumber wages, NOC 72300: <https://www.jobbank.gc.ca/wagereport/occupation/4754>
- Red Seal — Plumber: <https://red-seal.ca/eng/trades/plumber.shtml?wbdisable=true>

Colombia

- SENA Betowa — Instalación de redes hidrosanitarias: <https://betowa.sena.edu.co/oferta/instalacion-de-redes-hidrosanitarias?modality=P&offertype=company>
- SENA Betowa — Instalación de redes hidrosanitarias en PVC y griferías: <https://betowa.sena.edu.co/oferta/instalacion-de-redes-hidrosanitarias-en-pvc-y-griferias?modality=P&offertype=company&programId=216296>
- SENA Betowa — Básico de instalaciones sanitarias domiciliarias: <https://betowa.sena.edu.co/oferta/basico-de-instalaciones-sanitarias-domiciliarias?modality=P&offertype=company&programId=11816>
- SENA Betowa — Construcción de edificaciones: <https://betowa.sena.edu.co/oferta/construccion-de-edificaciones?modality=P&programId=195656>

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Authorship and responsible AI use

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